PROJECT REPORT

FEATURES AND TARGET SELECTION USING PYTHON

1. Abstract

This project focuses on selecting suitable input features and a target variable from a student performance dataset using Python, Pandas, and Scikit-learn.

The dataset contains academic information such as study hours, attendance percentage, assignment scores, previous examination scores, and final examination scores.

The project identifies relevant input features (X) and a target variable (y) for a supervised machine-learning problem.

It also demonstrates how a dataset can be divided into training and testing sets using Scikit-learn.

2. Introduction

Feature and target selection is an important step in supervised machine learning.

A supervised machine-learning problem uses input data and corresponding output values to learn patterns and make predictions.

Input features are the variables provided to the model, while the target variable is the output the model attempts to predict.

Selecting appropriate features helps provide meaningful information to a model and supports effective model development.

This project demonstrates feature and target selection using a student performance dataset.

3. Objectives

The main objectives are:

- To understand the basic structure of supervised machine-learning problems.
- To explore a student performance dataset.
- To identify numerical and categorical variables.
- To select suitable input features.
- To select an appropriate target variable.

- To explain why the selected variables are relevant.
- To understand the importance of avoiding target leakage.
- To demonstrate training and testing data separation.

4. Tools and Technologies

Python: Used as the primary programming language.

Pandas: Used to create, load, inspect, and select dataset columns.

Scikit-learn: Used for splitting the dataset into training and testing sets.

Google Colab: Used as the Jupyter Notebook development environment.

CSV: Used to store the dataset and selected feature and target information.

5. Dataset Description

The project uses a manually created student performance dataset containing 20 fictional student records.

The dataset consists of seven columns:

Column	Description
student_id	Unique student identifier
department	Academic department
study_hours	Hours spent studying
attendance_pct	Attendance percentage
assignment_score	Assignment marks
previous_exam_score	Previous examination marks
final_exam_score	Final examination marks

The dataset is created for educational purposes to demonstrate the selection of inputs and outputs in supervised machine learning.

6. Understanding Input Features and Target Variables

6.1 Input Features

Input features are the variables used by a machine-learning model to make predictions.

They are commonly represented by the variable X.

In this project, four input features are selected:

1. Study Hours
2. Attendance Percentage
3. Assignment Score
4. Previous Examination Score

These features provide information about the student's academic activities and previous performance.

6.2 Target Variable

The target variable is the output that the machine-learning model attempts to predict.

It is commonly represented by the variable y.

In this project, the selected target variable is:

Final Examination Score

The objective of a future supervised-learning model would be to predict final examination marks using the selected input features.

7. Feature Selection and Reasoning

The following features are selected:

Feature	Reason for Selection
study_hours	Represents time spent studying
attendance_pct	Represents student participation
assignment_score	Represents assignment performance
previous_exam_score	Represents previous academic performance

These features were selected because they provide relevant academic information that may help explain or predict examination performance.

8. Target Selection and Reasoning

The target variable is `final_exam_score`.

It is selected because the objective of the supervised-learning problem is to predict the student's final examination result.

The target variable is kept separate from the input features to ensure that the model does not receive the answer it is expected to predict.

9. Excluded Columns

Two columns are excluded from the selected numerical input features.

student_id: This is a unique identifier used to distinguish students. It does not provide meaningful academic performance information.

department: This is a categorical variable. It would need appropriate encoding before being included as a numerical model input. For this introductory feature-selection task, only numerical features are selected.

10. Feature and Target Separation

The input feature matrix is created using the selected columns:

X = study_hours, attendance_pct, assignment_score, previous_exam_score

The target variable is created separately:

y = final_exam_score

The resulting shapes are:

Component	Shape
Input features (X)	(20, 4)
Target variable (y)	(20,)

The input feature matrix contains 20 observations and four columns.

The target variable contains 20 corresponding output values.

11. Importance of Avoiding Data Leakage

Data leakage occurs when information that should not be available to a model during prediction is accidentally included in its input features.

In this project, `final_exam_score` is the target variable.

Therefore, it must not be included in X.

Keeping the target separate ensures that the selected inputs represent information intended to be available before the final examination result is predicted.

12. Training and Testing Data Separation

The dataset is divided into training and testing portions using Scikit-learn's `train_test_split()` function.

An 80:20 split is used.

Dataset	Percentage	Records
Training set	80%	16
Testing set	20%	4
Total	100%	20

The training data would be used to train a machine-learning model, while testing data would be used to evaluate its performance in a future modelling task.

This assignment demonstrates the split but does not train or evaluate a predictive model.

13. Methodology

The project follows these steps:

1) Import Pandas and Scikit-learn.
2) Create the student performance dataset.
3) Export and load the CSV dataset.
4) Inspect the dataset structure and column names.
5) Identify numerical and categorical columns.
6) Select four relevant input features.
7) Select the final examination score as the target.
8) Explain the reasoning for each selection.
9) Validate the feature and target separation.
10) Demonstrate the training and testing split.
11) Export the selected features, target, and summary files.

14. Applications

Feature and target selection is useful in:

- Student performance prediction.
- House price prediction.
- Employee salary prediction.
- Sales forecasting.
- Customer behavior prediction.
- Financial analysis.
- Healthcare prediction.
- Supervised machine-learning applications.

15. Limitations

The project uses a small, manually created dataset containing 20 fictional student records.

Only four numerical input features are selected for this introductory exercise.

The project does not train a machine-learning model or evaluate prediction accuracy.

The suitability of the selected features would need further examination using larger real-world datasets and additional domain information.

16. Future Enhancements

The project could be extended by:

- Using larger real-world student performance datasets.
- Encoding categorical variables such as department.
- Performing feature importance analysis.
- Applying feature selection algorithms.
- Training regression models.
- Evaluating model performance using appropriate metrics.
- Developing a complete student performance prediction system.

17. Conclusion

This project demonstrates the fundamental concepts of feature selection and target selection in supervised machine learning.

Four academic variables are selected as input features: study hours, attendance percentage, assignment score, and previous examination score.

The final examination score is selected as the target variable.

The project also demonstrates the separation of input and output data and the use of Scikit-learn to create training and testing datasets.

This exercise provides a foundation for understanding how datasets are prepared before developing machine-learning models.

18. References

Pandas Documentation: https://pandas.pydata.org/docs/

Scikit-learn Documentation: https://scikit-learn.org/stable/

Scikit-learn Train Test Split: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

Python Documentation: https://docs.python.org/3/

19. Project Links

GitHub Repository: https://github.com/venky1318/Features_and_Target_Selection